

High albite (C-1), BFDH: crystal morphology

Bravais–Friedel–Donnay–Harker (BFDH) morphology predicted from the high-albite lattice

Morphology of high albite predicted by the BFDH method from its C-1 lattice: each face is placed at a distance proportional to $1/d$, where d is the interplanar spacing. Because the lattice is C-centred, planes with $h + k$ odd are not lattice planes and their spacing is halved ($d_{\text{eff}} = d/2$). The listed forms are all of those that appear on the crystal.

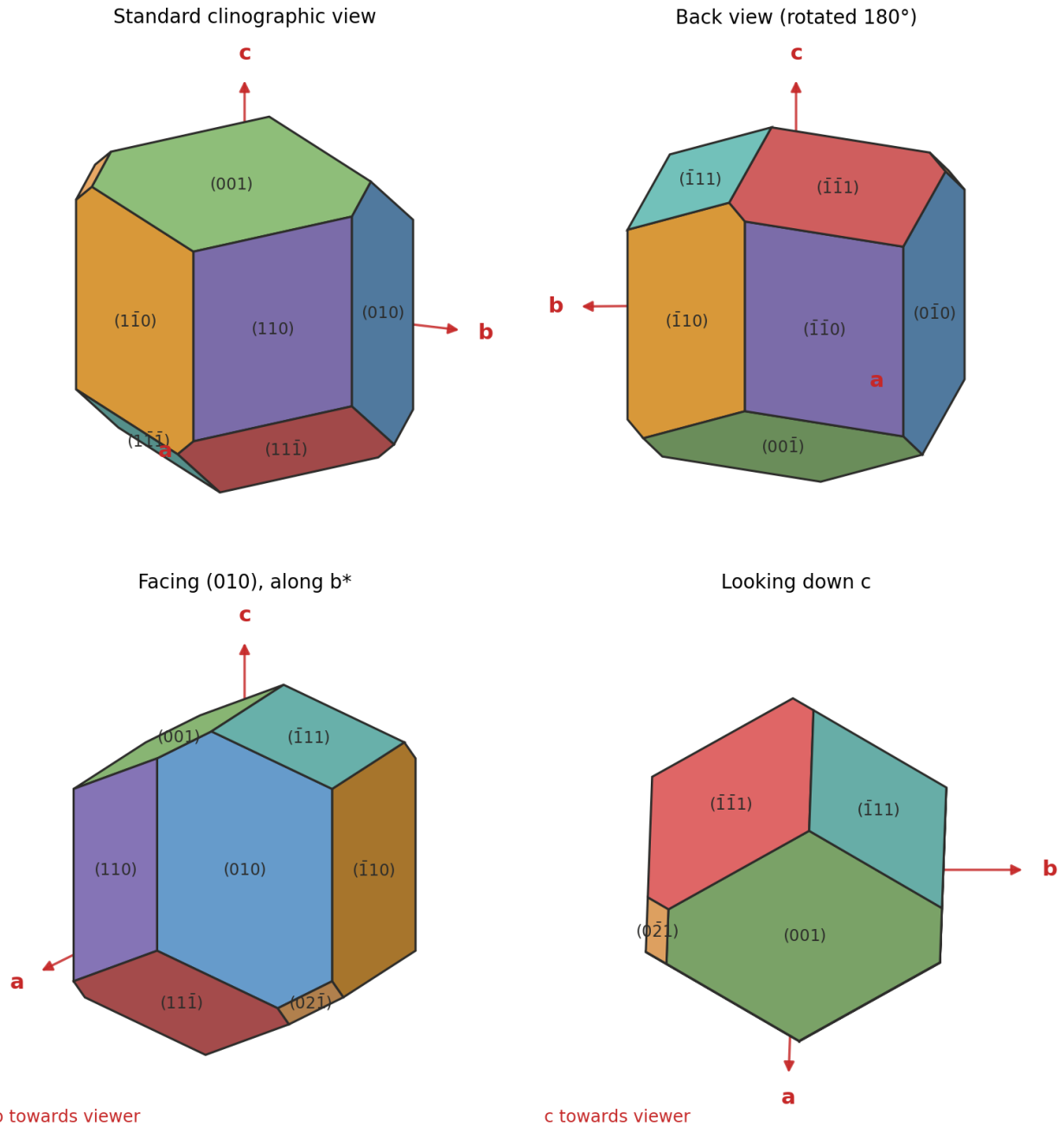


Figure 1. Orthographic views of the crystal in the frame used by SHAPE and the pymsh classes (c along z , b in the yz plane, centre at the origin). Red arrows are the crystal axes; faces with the same colour belong to the same form.

Crystal data

Quantity	Value
SHAPE file	HighAlbite_BFDH_C1bar.SHP (title HiAlbBFDH-C1bar, class code 2B1)
Unit cell a, b, c (Å)	8.16, 12.87, 7.11
α , β , γ (°)	93.5, 116.4, 90.3
Point group / morphology symmetry	-1 (triclinic, centrosymmetric)
Faces, corners, edges	14, 24, 36
Angle between c and the (001) normal	26.69°
pygmsh class	high_albite_bfdh (xtal_geometries_feldspars.py)

Forms

Form	d_eff (Å)	Central distance	Faces	Share of surface
{0 0 1}	6.353	1.01530	2	20.3%
{0 1 0}	6.419	1.00482	2	20.1%
{1 -1 0}	6.450	1.00000	2	18.9%
{1 1 0}	6.252	1.03163	2	15.6%
{-1 -1 1}	5.831	1.10616	2	13.8%
{-1 1 1}	5.649	1.14179	2	10.4%
{0 -2 1}	4.684	1.37700	2	0.9%

Central distances are those in the SHAPE file (arbitrary units). Surface shares are computed from the polyhedron built from these forms, which reproduces SHAPE's own corners to 10^{-7} .

Size and shape when the pygmsh class is called with z = 1

Quantity	Value
Scaling	$sf = z / fsp_z$ with $fsp_z = 1.262443$ (half-length along c in SHAPE units)
Length along c	2.000 (spans -1 to +1 along z)
Width (perpendicular to c, in (010))	1.849
Thickness (perpendicular to (010))	1.592
Length : width : thickness	1.26 : 1.16 : 1
Half-extents x , y , z	0.926, 0.813, 1.000
Volume V1	3.1982
Surface area	11.6285
Equivalent-sphere radius	0.9140
Surface / volume	3.636

To match a measured length L along c use $z = L/2$; to match a measured volume V use $z = (V / V1)^{1/3}$.

Source and notes

- Cell: a, b, c = 8.16, 12.87, 7.11 Å; α , β , γ = 93.5, 116.4, 90.3°; Z = 4; space group C-1. The β supplied (12.87°) repeats the value of b; 116.4° (the published high-albite value) was used.
- Distances are $1/d_{\text{eff}}$, normalised so that the nearest face, (1 -1 0), is at 1.000.
- BFDH uses lattice geometry only (no bonding or growth conditions), so it is a first estimate of habit. Natural albite is often more tabular on (010).
- A monoclinic C2/m (monalbite) version was also computed earlier; it has equal (110)/(1-10) and (-1 1 1)/(-1-1 1) faces and 12 faces instead of 14.
- Scaling in the pygmsh class: $fsp_z = 1.262443$, the half-length along c.

SHAPE input file (crystal definition)

```
HiAlbBFDH-C1bar
2B1      0      1  3.10      0
8.1600000 12.8700000 7.1100000
93.5000000 116.4000000 90.3000000
1      -1      0      1.00000  0  0  0  0
0      1      0      1.00482  0  0  0  0
0      0      1      1.01530  0  0  0  0
1      1      0      1.03163  0  0  0  0
-1     -1      1      1.10616  0  0  0  0
-1      1      1      1.14179  0  0  0  0
0      -2      1      1.37700  0  0  0  0
99      99      99      1.00000  0  0  0  0
```

Line 1: title. Line 2: crystal class code (2 = triclinic -1) and SHAPE options. Lines 3-4: a, b, c and alpha, beta, gamma. Following lines: h, k, l and central distance of each form, ending with 99 99 99. The display settings that follow in the file are not shown.